# Prodigy Anesthesia Quick Review

## Herbal Medications

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| Approximately what percentage of the US population takes herbal medications? What percentage of these report to healthcare providers that they take herbal medications? | Over 35% of the populations takes some form of herbal medication, but only 30% of this population accurately reports that they take them when asked by a healthcare provider.<br><br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 315.* |
| Does the FDA regulate herbal medications? | The FDA does not regulate herbal supplements as they do other medications. The FDA can, however, remove an herbal supplement from the market if they find that it is not fit for human consumption (for example if it is contaminated with toxic pesticides or fertilizers or if it is not manufactured according to standard practices).<br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 316.* |
| What legislative act classified herbal medications as dietary supplements rather than medications? | The U.S. Dietary Supplement Health and Education Act defined herbal medications as dietary supplements rather than regulated medications.<br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 316.* |
| There are eleven common herbal supplements that have significant anesthetic implications. Name them. | Valerian, St. John's wort, saw palmetto, kava, ginseng, ginkgo biloba, ginger, garlic, feverfew, ephedra, and echinacea<br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 317.* |
| Several herbal medications promote bleeding. Which one does so by inhibiting platelet-activating factor? | Gingko biloba<br><br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 317.* |
| Several herbal medications promote bleeding. Which one does so by promoting fibrinolysis? | Garlic<br><br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 317.* |
| Several herbal medications promote bleeding. Which one does so by inhibiting cyclooxygenase? | Saw palmetto<br><br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 317.* |

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| What herbal supplement is known to be metabolized by the cytochrome p450 pathway? | 50% of St John's wort is metabolized via the cytochrome p450 pathway<br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 317.* |
| What herbal medication should be tapered over 7 days to prevent withdrawal symptoms? | Valerian, which potentiates barbiturates, should be tapered over 7 days to prevent withdrawal symptoms<br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 317.* |
| Name three herbal medications that have been shown to potentiate barbiturates. | Kava, saw palmetto, and valerian<br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 317.* |
| Name five herbal medications that have been shown to increase the risk of bleeding. | Feverfew, garlic, ginger, ginkgo biloba, and ginseng<br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 317.* |
| For what reasons is echinacea commonly used and what are its most significant side effects? | Echinacea is taken for the treatment and prevention of upper respiratory infections and flu. Although it has not been proven conclusively effective in the prevention of colds or flu, it has demonstrated some degree of immunostimulation and should be avoided before transplant surgery. It may also result in increased asthma symptoms and exacerbate allergic rhinitis. In one study, pediatric patients exposed to echinacea were significantly more likely to develop a rash than patients who had not been exposed to it.<br>*Katzung BG, Masters, SB, & Trevor AJ. Basic and Clinical Pharmacology. 12th ed. New York: McGraw-Hill; 2012: 1128.* |
| Which herbal medication may result in lowered blood glucose levels? | Ginseng<br>*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 346.* |
| In general, how long should herbal medications be discontinued prior to anesthesia? | 1-2 weeks. Patients should be instructed to taper these medications to prevent withdrawal symptoms that can occur from herbal medications like valerian.<br>*Ouellette RG and Joyce JA. Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones & Bartlett Learning; 2011: 316.* |

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How would the use of St John's wort affect serum digoxin levels?

Because St. John's wort causes enzyme induction, it can result in decreased serum digoxin levels. It can also similarly affect warfarin, calcium channel blockers, protease inhibitors, benzodiazepines, and many other drugs.

*Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 346.*

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